BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 209 · Exam-style data representation problems

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does CountAtLeast([5,7,3],5) return?

Q2. What does O(log n) often describe in binary search?

Q3. What is the sign bit for a negative 8-bit two's-complement integer?

Q4. Calculate $0.1000_2 \times 2^3$.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

bit width — Number of bits in a representation.

two's complement — Fixed-width signed binary format.

mantissa — Signed fractional significant component.

exponent — Power-of-two scaling value.

precision — Number of significant digits retained.

New learning

Respect the bit width — Write positive 5 using exactly eight bits.

Invert then add one — The result is 11111011_2, representing −5 in 8-bit two's complement.

Interpret floating-point differently — 0.1100_2 is fractional 0.75; multiplying by 2^3 gives 6.

Worked example

Convert −10 to 8-bit two's complement.

1. Predict — Write 00001010 for +10 and invert to 11110101.

2. Explain — Add one: 11110110 represents −10.

We Do · Practise with guidance

Convert −7 to 8-bit two's complement.

```
+7=00000111
NOT=?
+1=?
```

Predict

Start with +7 = 00000111 and invert all bits.

Explain

Add one to 11111000, giving 11111001 for −7.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Convert unsigned binary 00001011 to decimal.

Question 2

Encode −12 in 8-bit two's complement.

Question 3

Normalise positive $0.0011_2 \times 2^5$ using a two's-complement fractional mantissa.

Exit Ticket · Answer before checking

Eight-bit two's complement can represent decimal +128.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 2.

Q2. Repeatedly halving the candidate interval.

Q3. 1.

Q4. 4.

You Do pages · Independent questions

Q1. 11.

Q2. +12=00001100, inverted 11110011, add1 = 11110100.

Q3. $0.0011\times2^{5} \rightarrow 0.0110\times2^{4} \rightarrow 0.1100\times2^{3}$. Numerical value 6.

Exit Ticket · Model answer

False. Its integer range is −128 to +127.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: